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Patent Public Search | Text View

United States Patent Application Publication

20250278276

Kind Code

A1

Publication Date

September 04, 2025

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APPARATUSES, SYSTEMS, AND METHODS FOR MODE REGISTER PAGE ACCESS MODES

Abstract

Embodiments disclosed herein provide techniques for entering, remaining, and exiting a mode register page access (MRPA) mode in a semiconductor device (e.g., a memory device). A command is used to cause the memory device to enter the MRPA mode when accessing the mode register, and a subsequent command is used to cause the memory device to remain in the MRPA mode for continued access to the mode register. While in the MRPA mode, subsequent commands enable the memory device to be accessed using a mode register address for a read command, or a mode register address and an operation code (OP code) for a write command.

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Appl. No.: 19/032886

Filed: January 21, 2025

Related U.S. Application Data

us-provisional-application US 63561159 20240304

Publication Classification

Int. Cl.: G06F9/30 (20180101); G06F9/355 (20180101)

U.S. Cl.:

CPC G06F9/30189 (20130101); G06F9/355 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the filing benefit of U.S. Provisional Application No. 63/561,159, filed Mar. 4, 2024. The Provisional Application is incorporated by reference herein in its entirety and for all purposes.

BACKGROUND

[0002] A semiconductor memory device may include a number of mode registers which are used to store information, such as configuration and status information in the form of operation (OP) codes represented by binary digits (or “bits”). Typically, in a dynamic random access memory (DRAM), when information in a mode register is written to or read from a particular address in the mode register, a full command is transmitted to the semiconductor memory device. For a mode register read operation, the full command includes the command and the mode register address. For a mode register write operation, the full command includes the command, the mode register address, and the OP code to be written to the mode register. As the number of mode registers in a semiconductor memory device increases, the requirement for the full command data for each read and write operation can result in slower mode register read and write operations.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Non-limiting and non-exhaustive examples are described with reference to the following Figures. The elements of the drawings are not necessarily to scale relative to each other. Identical reference numerals have been used, where possible, to designate identical features that are common to the Figures.

[0004] FIG. 1 illustrates a block diagram of an example system according to an embodiment of the disclosure;

[0005] FIG. 2 illustrates a block diagram of a semiconductor device according to an embodiment of the disclosure;

[0006] FIG. 3 illustrates a flowchart of an example method of operating a semiconductor device according to an embodiment of the disclosure;

[0007] FIG. 4 illustrates a first example of a command truth table according to an embodiment of the disclosure;

[0008] FIG. 5 illustrates a second example of a command truth table according to an embodiment of the disclosure;

[0009] FIG. 6A illustrates a first technique for entering and remaining in the mode register page access mode according to an embodiment of the disclosure;

[0010] FIG. 6B illustrates a first technique for exiting the mode register page access mode according to an embodiment of the disclosure;

[0011] FIG. 6C illustrates a second technique for entering or exiting the mode register page access mode according to an embodiment of the disclosure; and

[0012] FIG. 6D illustrates a third technique for entering or exiting the mode register page access mode according to an embodiment of the disclosure.

DETAILED DESCRIPTION

[0013] Embodiments disclosed herein provide techniques for entering, remaining, and exiting a mode register page access (MRPA) mode in a semiconductor device (e.g., a memory device). In some embodiments of the disclosure, the MRPA mode refers to the execution or the performance of multiple MRPA write operations to the same page or MRPA read operations from the same page. A command is used to cause the memory device to enter the MRPA mode when accessing the mode register, and a subsequent command is used to cause the memory device to remain in the MRPA mode for continued access to the mode register. While in the MRPA mode, subsequent commands enable the memory device to be accessed using a mode register address for a read command, or a mode register address and an operation code (OP code) for a write command.

[0014] FIG. 1 illustrates a block diagram of an example system **100** according to an embodiment of the disclosure. The system **100** includes a controller **10** and a memory system **105**. The memory system **105** includes memories **110(0)-110(p)** (e.g., “Device 0” through “Device p”), where p is a number of the memories. The memories **110(0)-110(p)** may be dynamic random access memory (DRAM). The memories **110(0)-110(p)** may be double data rate (DDR), low power double data rate (LPDDR), graphics double data rate (GDDR), or other DRAM in some embodiments of the disclosure. The memories **110(0)-110(p)** are each coupled to command/address, data, and clock busses. The controller **10** and the memory system **105** are in communication over several busses. Commands and addresses CA are received by the memory system **105** on a command/address bus **115**, and data DQ is provided between the controller **10** and the memory system **105** over a data bus **125**. Various clocks may be provided between the controller and memory system **105** over a clock bus **130**. The clock bus **130** may include signal lines for providing system clocks CK_t and CK_c received by the memory system **105** and data clocks (strobes) DQS_t and DQS_c received and/or provided by the memory system **105** to the controller **10**. Each of the busses may include one or more signal lines on which signals are provided.

[0015] The CK_t and CK_c clocks provided by the controller **10** to the memory system **105** are used for timing the provision and receipt of the commands and addresses. The DQS_t and DQS_c clocks are used for timing the provision of data. The CK_t and CK_c clocks are complementary, and the DQS_t and DQS_c clocks are complementary. Clocks are complementary when a rising edge of a first clock occurs at a same time as a falling edge of a second clock, and when a rising edge of the second clock occurs at a same time as a falling edge of the first clock.

[0016] The controller **10** provides commands to the memory system **105** to perform memory operations. Examples of memory commands include timing commands for controlling the timing of various operations, access commands for accessing the memory, such as read commands for performing read operations and write commands for performing write operations, mode register write and read commands for performing mode register write and read operations, as well as other commands and operations. The command signals provided by the controller **10** to the memory system **105** further include external control signals (e.g., CS_n(0), CS_n(1), CS_n(p)). While all of the memories **110(0)-110(p)** are provided the commands, addresses, data, and clocks, the external control signals provided on respective select signal lines are used to select which of the memories **110(0)-110(p)** will respond to the command and perform the corresponding operation. In some embodiments of the disclosure, a respective control signal is provided to each memory **110(0)-110(p)** of the memory system **105**. The controller **10** provides an active control signal to select the corresponding memory **110(0)-110(p)**. While the respective control signal is active, the corresponding memory **110(0)-110(p)** is selected to receive the commands and addresses provided on the command/address bus **115**. In some embodiments of the disclosure, the external control signal is used in combination with the CA signals to indicate different memory commands and memory operations.

[0017] In operation, when a read command and associated address are provided by the controller **10** to the memory system **105**, the memory **110(0)-110(p)** selected by the external control signals

receives the read command and associated address and performs a read operation to provide the controller **10** with read data from a memory location corresponding to the associated address. The read data is provided by the selected memory **110(0)-110(p)** to the controller **10** according to a timing relative to receipt of the read command. For example, the timing may be based on a read latency (RL) value that indicates the number of clock cycles of the CK_t and CK_c clocks (a clock cycle of the CK_t and CK_c clocks is referenced as tCK) after the read command when the read data is provided by the selected memory **110(0)-110(p)** to the controller **10**. The RL value is programmed by the controller **10** in the memories **110(0)-110(p)**. For example, the RL value may be programmed in respective mode registers of the memories **110(0)-110(p)**. As known, mode registers included in each of the memories **110(0)-110(p)** may be programmed with information for setting various operating modes and/or to select features for operation of the memories. One of the settings may be for the RL value.

[0018] In preparation of the selected memory **110(0)-110(p)** providing the read data to the controller **10**, the memory **110(0)-110(p)** provides active data clocks DQS_t and DQS_c. A clock is active when the clock transitions between low and high clock levels periodically. Conversely, a clock is inactive when the clock maintains a constant clock level and does not transition periodically. The DQS_t and DQS_c clocks are provided by the memory **110(0)-110(p)** performing the read operation to the controller **10** for timing the provision of read data to the controller **10**. The controller **10** may use the DQS_t and DQS_c clocks for receiving the read data.

[0019] In operation, when a write command and associated address are provided by the controller **10** to the memory system **105**, the memory **110(0)-110(p)** selected by the external control signals receives the write command and associated address and performs a write operation to write data from the controller **10** to a memory location corresponding to the associated address. The write data is provided to the selected memory **110(0)-110(p)** by the controller **10** according to a timing relative to receipt of the write command. For example, the timing may be based on a write latency (WL) value that indicates the number of clock cycles of the CK_t and CK_c clocks after the write command when the write data is provided to the selected memory **110(0)-110(p)** by the controller **10**. The WL value is programmed by the controller **10** in the memories **110(0)-110(p)**. For example, the WL value may be programmed in respective mode registers of the memories **110(0)-110(p)**.

[0020] In preparation of the selected memory **110(0)-110(p)** receiving the write data from the controller **10**, the controller **10** provides active data clocks DQS_t and DQS_c to the memory system **105**. The DQS_t and DQS_c clocks may be used by the selected memory **110(0)-110(p)** to generate internal clocks for timing the operation of circuits to receive the write data. The data is provided by the controller **10** and the selected memory **110(0)-110(p)** receives the write data according to the DQS_t and DQS_c clocks, which is written to memory corresponding to the memory addresses.

[0021] Mode register write commands and mode register read commands can be used to access the mode registers (e.g., mode register **230** in FIG. 2). In operation, when a mode register read command and associated address are provided by the controller **10** to the memory system **105**, the mode register selected by the select signals receives the read command and associated address and performs a read operation to provide the controller **10** with read data from the mode register corresponding to the associated address. The read data is provided by the selected mode register to the controller **10**. When a mode register write command and associated address are provided by the controller **10** to the memory system **105**, the mode register selected by the select signals receives the write command and associated address and performs a write operation to write data from the controller **10** to a mode register corresponding to the associated address. The write data is provided to the selected mode register by the controller **10**.

[0022] FIG. 2 illustrates a block diagram of a semiconductor device **200** according to an embodiment of the disclosure. The semiconductor device **200** may include, without limitation, a DRAM. The semiconductor device **200** includes a memory array **250**. The memory array **250** is

shown as including a plurality of memory banks. In the embodiment of FIG. 2, the memory array **250** is shown as including memory banks BANK0-BANKm. Each memory bank includes a plurality of word lines WL, a plurality of bit lines BL and/BL, and a plurality of memory cells MC arranged at intersections of the plurality of word lines WL and the plurality of bit lines BL and/BL. Selection of the word line WL is performed by a row decoder **240** and selection of the bit lines BL and/BL is performed by a column decoder **245**. In the embodiment of FIG. 2, the row decoder **240** includes a respective row decoder for each memory bank and the column decoder **245** includes a respective column decoder for each memory bank. The bit lines BL and/BL are coupled to a respective sense amplifier (SAMP). Read data from the bit line BL or/BL is amplified by the sense amplifier SAMP and transferred to read/write amplifiers **255** over complementary local data lines (LIOT/B), transfer gate (TG), and complementary main data lines (MIOT/B). Conversely, write data outputted from the read/write amplifiers **255** is transferred to the sense amplifier SAMP over the complementary main data lines MIOT/B, the transfer gate TG, and the complementary local data lines LIOT/B, and written in the memory cell MC coupled to the bit line BL or/BL.

[0023] A mode register **230** stores information, for example, configuration and status information for the semiconductor device **200**. The mode register **230** may be accessed through mode register read commands and mode register write commands. The mode register access commands cause the semiconductor device **200** to perform mode register read operations and mode register write operations. A mode register read command causes the semiconductor device **200** to provide information stored by the mode register that is accessed, and a mode register write command causes the semiconductor device **200** to store information in the mode register that is accessed. The mode register **230** may include several mode registers, with each of the mode registers corresponding to a mode register address and storing different types of information. In some embodiments of the disclosure, one or more mode registers may be grouped into a mode register page (MRP). The group of mode registers can be grouped by the types of information stored, such as storing information having similar characteristics, such as information related to modes of operation, input/output related configuration settings, various calibration settings, as well as others.

[0024] The semiconductor device **200** may employ a plurality of external terminals that include command and address and control terminals (CA0-CAn and CS_n) coupled to a command and address bus to receive commands and addresses, and an external control CS_n signal. The external terminals may further include clock terminals to receive clocks CK_t and CK_c, and data clocks DQS_t and DQS_c, data terminals DQ and DM, and power supply terminals to receive power supply potentials VDD, VSS, and VDDQ.

[0025] The clock terminals are supplied with external clocks CK_t and CK_c that are provided to an input buffer **220**. The external clocks may be complementary. The input buffer **220** generates an internal clock ICLK based on the CK_t and CK_c clocks. The ICLK clock is provided to the command decoder **215** and to an internal clock generator **222**. The internal clock generator **222** provides various internal clocks LCLK based on the ICLK clock. The LCLK clocks may be used for timing operations of various internal circuits.

[0026] The CA terminals (e.g., CA0-CAn) may be supplied with memory addresses. The memory addresses supplied to the CA terminals are transferred, via a command/address input circuit **205**, to an address decoder **212**. The address decoder **212** receives the address and supplies a decoded row address XADD to the row decoder **240** and supplies a decoded column address YADD to the column decoder **245**. The CA terminals may be supplied with commands. Examples of commands include access commands for accessing the memory, such as read commands for performing read operations and write commands for performing write operations, mode register write and read commands for performing mode register write and read operations, as well as other commands and operations.

[0027] The commands may be provided as internal command signals to a command decoder **215** via the command/address input circuit **205**. The command decoder **215** includes circuits to decode

the internal command signals to generate various internal signals and commands for performing operations. For example, the command decoder **215** may provide a row command signal ACT to select a word line and a column command signal R/W to select a bit line.

[0028] The power supply terminals are supplied with power supply potentials VDD and VSS. The power supply potentials VDD and VSS are supplied to an internal voltage generator circuit **270**. The internal voltage generator circuit **270** generates various internal potentials VPP, VOD, VARY, VPERI, and the like based on the power supply potentials VDD and VSS supplied to the power supply terminals. The internal potential VPP is mainly used in the row decoder **240**, the internal potentials VOD and VARY are mainly used in the sense amplifiers SAMP included in the memory array **250**, and the internal potential VPERI is used in many peripheral circuit blocks.

[0029] The power supply terminals are also supplied with power supply potentials VDDQ and VSS. The power supply potentials VDDQ and VSS are supplied to the input/output circuit **260**. The power supply potentials VDDQ and VSS supplied to the power supply terminals may be the same potentials as the power supply potentials VDD and VSS supplied to the power supply terminals in an embodiment of the disclosure. The power supply potentials VDDQ and VSS supplied to the power supply terminals may be different potentials from the power supply potentials VDD and VSS supplied to the power supply terminals in another embodiment of the disclosure. The power supply potentials VDDQ and VSS supplied to the power supply terminals are used for the input/output circuit **260** so that power supply noise generated by the input/output circuit **260** does not propagate to the other circuit blocks.

[0030] When a read command is received, and a row address and a column address are timely supplied with the read command, read data is read from memory cells in the memory array **250** corresponding to the row address and column address. The read command is received by the command decoder **215**, which provides internal commands so that read data from the memory array **250** is provided to the read/write amplifiers **255**. The read data is output to outside from the data terminals DQ via the input/output circuit **260**. The DQS_t and DQS_c clocks are provided externally from clock terminals for timing provision of the read data by the input/output circuit **260**. The external terminals DQ include several separate terminals, each providing a bit of data synchronized with a clock edge of the DQS_t and DQS_c clocks.

[0031] When a write command is received, and a row address and a column address are timely supplied with the write command, write data supplied to the data terminals DQ is written to a memory cells in the memory array **250** corresponding to the row address and column address. A data mask may be provided to the data terminals DM to mask portions of the data when written to memory. The write command is received by the command decoder **215**, which provides internal commands so that the write data is received by input receivers in the input/output circuit **260**. DQS_t and DQS_c clocks are also provided to the external clock terminals (e.g., by a controller) for timing the receipt of the write data by the input receivers of the input/output circuit **260**. The write data is supplied via the input/output circuit **260** to the read/write amplifiers **255**, and by the read/write amplifiers **255** to the memory array **250** to be written into a memory cell MC. As previously described, the external terminals DQ include several separate terminals. With reference to a write operation, each external terminal DQ concurrently receives a bit of data synchronized with a clock edge of the DQS_t and DQS_c clocks.

[0032] When a mode register read command is received, and a mode register address is timely supplied with the mode register read command, read data is read from the mode register **230** corresponding to the mode register address. The read data is output to outside from the data terminals DQ via the input/output circuit **260**. When a mode register write command is received, and a mode register address is timely supplied with the mode register write command, write data supplied to the data terminals DQ is written to the mode register **230** corresponding to the mode register address.

[0033] FIG. **3** illustrates a flowchart of an example method **300** of operating a semiconductor

device according to an embodiment of the disclosure. In some embodiments of the disclosure, the method may be performed by the semiconductor device **200** of FIG. 2 and/or the memories **110(0)-110(p)** of FIG. 1.

[0034] At block **302**, a command to perform a mode register read operation or a mode register write operation is received. A determination is made at block **304** as to whether a mode register read command associated with the mode register read operation, or a mode register write command associated with the mode register write operation, causes the semiconductor device (e.g., causes one or more mode registers in the semiconductor device) to enter an MRPA mode. In some embodiments of the disclosure, the MRPA mode refers to the execution or the performance of multiple MRPA write operations to the same page or MRPA read operations from the same page. Non-limiting nonexclusive command truth tables for mode register page access (MRPA) read commands and MRPA write commands associated with the MRPA mode are described in more detail in conjunction with FIG. 4 and FIG. 5.

[0035] If a determination is made at block **304** that the mode register read command or the mode register write command does not cause the semiconductor device (e.g., cause one or more mode registers in the semiconductor device) to enter the MRPA mode, the method passes to block **306** where a mode register read operation defined by the mode register read command or a mode register write operation defined by the mode register write command is performed. The mode register read command and the mode register write command are direct mode register access commands. With direct mode register access commands, a mode register corresponding to the mode register address is accessed once. For subsequent mode register read operations, the full mode register read command is received for each mode register read operation (e.g., the command, the mode register page address, and the mode register address). For subsequent mode register write operations, the full mode register write command is received for each mode register write operation (e.g., the command, the mode register page address, the mode register address, and the OP code to be written to the mode register). Non-limiting nonexclusive command truth tables for direct mode register read commands and direct mode register write commands are described in more detail in conjunction with FIG. 4 and FIG. 5.

[0036] When a determination is made at block **304** that the mode register read command or the mode register write command will cause the semiconductor device (e.g., cause one or more mode registers in the semiconductor device) to enter the MRPA mode, the method continues at block **308** where the semiconductor device enters the MRPA mode. The mode register read command and the mode register write command that cause the semiconductor device to enter the MRPA mode are MRPA commands. With MRPA commands, a mode register page corresponding to a mode register page address is identified and is available for multiple accesses to that mode register page. For example, an MRPA write command causes a write operation associated with the MRPA write command to be performed and enables subsequent write operations to be performed on the same mode register page until the semiconductor device exits the MRPA mode (or until the last MRPA write command is received). An MRPA read command causes a read operation associated with the MRPA read command to be performed and enables subsequent read operations to be performed on the same mode register page until the semiconductor device exits the MRPA mode (or until the last MRPA read command is received).

[0037] Various techniques can be used to cause the semiconductor device to enter the MRPA mode. In one embodiment, an external control signal can toggle to a particular state in the MRPA read command or in the MRPA write command. In one embodiment, the external control signal is a CS signal. For example, the external control signal may toggle to a low state for one clock cycle (e.g., a system clock provided to a semiconductor device) to enter the MRPA mode. A non-limiting nonexclusive example of the use of the external control signal to cause the semiconductor device to enter the MRPA mode is described in more detail in conjunction with FIGS. 4-6A.

[0038] In another embodiment, one or more command address (CA) bits can be set to a particular

state in the MRPA read command or in the MRPA write command to cause the semiconductor device to enter the MRPA mode. For example, the last CA bit in the sequence of CA bits may be set to a low state to cause the semiconductor device to enter the MRPA mode. A non-limiting nonexclusive example of the use of the CA bits to cause the semiconductor device to enter the MRPA mode is described in more detail in conjunction with FIG. 6D.

[0039] In yet another embodiment, some or all of the mode register address (MRA) bits can be set at a particular state or a particular sequence of states in the mode register page access read command or in the mode register page access write command to cause the semiconductor device to enter the MRPA mode. For example, in an embodiment when there are six MRA bits, all of the six MRA bits may be set to a low state to cause the semiconductor device to enter the MRPA mode. A non-limiting nonexclusive example of the use of the MRA bits to cause the semiconductor device to enter the MRPA mode is described in more detail in conjunction with FIG. 6C.

[0040] Next, as shown in block **310**, a determination is made as to whether an MRPA read command or an MRPA write command was received at block **302**. If a determination is made that an MRPA read command was received, the method proceeds at block **312** where the MRPA read operation is performed in the mode register page corresponding to the MRP of the MRPA read command, and in the mode register corresponding to the MRA of the MRPA read command. Additional mode registers in the same MRP are read in the MRPA mode when one or more additional MRPA read commands including another MRA are received, causing one or more corresponding additional read operations to be performed in the same MRP. Generally, an MRPA read command causes the semiconductor device to enter the MRPA mode and establishes the MRP and a first mode register of the MRP to be accessed. Each subsequent MRPA read command includes an MRA in the same MRP to be read. Additional MRPA read operations for mode registers in the same MRP continue until the semiconductor device exits the MRPA mode at block **314**.

[0041] Returning to block **310**, when a determination is made that an MRPA write command was received, the method passes to block **316**. At block **316**, the MRPA write operation is performed in a mode register page corresponding to the MRP of the MRPA write command, and in the mode register corresponding to the MRA of the MRPA write command. Additional mode registers in the same MRP are programmed in the MRPA mode when one or more additional MRPA write commands including another MRA are received, causing one or more additional write operations to be performed in the same MRP. Generally, an MRPA write command causes the semiconductor device to enter the MRPA mode and establishes the MRP and a first mode register of the MRP to be accessed. Each subsequent MRPA write command includes an MRA to be written to in the same MRP and the OP code for the mode register write operation. Additional MRPA write operations for mode registers in the same MRP continue until the semiconductor device exits the MRPA mode at block **314**.

[0042] Various techniques may be used to cause the semiconductor device (e.g., cause one or more mode registers in the semiconductor device) to exit the MRPA mode. In one embodiment, the external control signal can toggle to a particular state in the MRPA read command or in the MRPA write command for multiple consecutive clock cycles (e.g., consecutive rising edges or falling edges). For example, the external control signal may toggle to a low state for two consecutive clock cycles to cause the semiconductor device to exit the MRPA mode. A non-limiting nonexclusive example of toggling the state of the external control signal is described in more detail in conjunction with FIG. 6B.

[0043] In another embodiment, one or more CA bits in the MRPA read command or in the MRPA write command can be set to a particular state to cause the semiconductor device to exit the MRPA mode. For example, the last CA bit in the sequence of CA bits may be set to a low state to cause the semiconductor device (e.g., cause one or more mode registers in the semiconductor device) to exit the MRPA mode. A non-limiting nonexclusive example of the use of the one or more CA bits to exit the MRPA mode is described in more detail in conjunction with FIGS. 4, 5, and 6D.

[0044] In yet another embodiment, some or all of the MRA bits can be set at a particular state or to a particular sequence of states in the MRPA read command or in the MRPA write command. For example, in an embodiment when there are six MRA bits, all of the six mode register bits may be set to a low state to cause the mode register to exit the MRPA mode. A non-limiting nonexclusive example of the use of the mode register address bits to cause the semiconductor device (e.g., cause one or more mode registers in the semiconductor device) to exit the MRPA mode is described in more detail in conjunction with FIG. 6C.

[0045] FIG. 4 illustrates a first example of a command truth table **400** according to an embodiment of the disclosure. The example command truth table **400** includes an example configuration for a 7-bit CA bus. Although the illustrated command truth table **400** depicts low (L), high (H), and valid (V) states in some fields, other embodiments are not limited to these states or to the illustrated sequence of states. Additionally, or alternatively, a command may or may not include command bits (e.g., see the MRPA write extension same page command **412** in FIGS. 4 and 512 in FIG. 5). In some embodiments, the command truth table **400** may be implemented with the semiconductor device **200** of FIG. 2 and/or the memories **110(0)-110(p)** of FIG. 1. In some embodiments, the command truth table **400** may be implemented with the method **300** of FIG. 3.

[0046] The command truth table **400** includes column entries for a command (Function) **402**, the external control signal (CS_n) **404**, and the CA bits (CA0-CA6) **406**. The example command truth table **400** includes row entries for each command **402**. The commands **402** include a mode register write command **408**, an MRPA write same page command **410**, an MRPA write extension same page command **412**, a mode register read command **414**, an MRPA read same page command **416**, and an MRPA read extension same page command **418**.

[0047] The mode register write command **408** is a 4-cycle command in which each sub-row **420**, **422**, **424**, **426** is received at one clock cycle (for a total of four clock cycles for the entire command). The mode register write command **408** is an example of a write command that is performed at block **306** in FIG. 3. The MRPA write same page command **410** is a 4-cycle command in which each sub-row **428**, **430**, **432**, **434** is received at one clock cycle (for a total of four clock cycles for the entire command). The MRPA write extension same page command **412** is a 2-cycle command in which each sub-row **436**, **438** is received at one clock cycle (for a total of two clock cycles for the entire command). The MRPA write same page command **410** and the MRPA write extension same page command **412** are example write commands that are performed at block **316** in FIG. 3.

[0048] The mode register read command **414** is a 3-cycle command in which each sub-row **440**, **442**, **444** is received at one clock cycle (for a total of three clock cycles for the entire command). The mode register read command **414** is an example read command that is performed at block **306** in FIG. 3. The MRPA read same page command **416** is a 3-cycle command in which each sub-row **446**, **448**, **450** is received at one clock cycle (for a total of three clock cycles for the entire command). The MRPA read extension same page command **418** is a 1-cycle command in which the sub-row **452** is received at one clock cycle. The MRPA read same page command **416** and the MRPA read extension same page command **418** are example read commands that are performed at block **312** in FIG. 3.

[0049] The mode register write command **408** can be used for standalone mode register write operations (e.g., when the semiconductor device is not in the MRPA mode). The mode register write command **408** is an example of a direct mode register command. The sequence of the states (H or L) in the CA0-CA4 bits in sub-row **420** define the command to be performed, which is the mode register write command **408**. The sequence of the states in MRP0-MRP4 in the CA0-CA4 bits in sub-row **422** identify the MRP to be accessed. The sequence of the states in MRA0-MRA5 in the CA0-CA5 bits in sub-row **424** define the MRA in the MRP to be accessed. The state in OP0 in the CA6 bit in sub-row **424** along with the sequence of the states in OP1-OP7 in the CA0-CA6 bits in sub-row **426** provide the OP code for the write operation.

[0050] The MRPA write same page command **410** can be used to cause the semiconductor device (e.g., cause one or more mode registers in the semiconductor device) to enter the MRPA mode. In the illustrated embodiment, the external control signal **404** is used to cause the semiconductor device to enter the MRPA mode. The external control signal **404** toggles, or is set to, a low state in sub-row **432** (compared to the state of the external control signal **404** in sub-row **424**), which indicates the semiconductor device is to enter the MRPA mode. This technique is shown and described in more detail in conjunction with FIG. 6A. Other techniques for causing the semiconductor device (e.g., causing one or more mode registers in the semiconductor device) to enter the MRPA mode are described in more detail in conjunction with FIGS. 6C-6D.

[0051] In the MRPA write same page command **410**, the sequence of the states (H or L) in the CA0-CA4 bits in sub-row **428** define the command to be performed, which is the MRPA write same page command **410**. The sequence of the states in MRP0-MRP4 in the CA0-CA4 bits in sub-row **430** provide the MRP address to be accessed. The sequence of the states in MRA0-MRA5 in the CA0-CA5 bits in sub-row **432** define the MRA in the MRP to be accessed. The state in OP0 in the CA6 bit in sub-row **432** along with the sequence of the states in OP1-OP7 in the CA0-CA6 bits in sub-row **434**, respectively, provide the OP code for the write operation.

[0052] When one or more additional MRPA write operations are performed on the same MRP, the MRPA write extension same page command **412** is used to cause the semiconductor device (e.g., cause one or more mode registers in the semiconductor device) to remain in the MRPA mode and perform an additional MRPA write operation on the same page. The MRPA write extension same page command **412** is received for each additional MRPA write operation to be performed. In the example embodiment, the external control signal **404** toggles, or is set to, a low state in sub-row **436**, which indicates the semiconductor device is to remain in the MRPA mode. The sequence of the states in MRA0-MRA5 in the CA0-CA5 bits in sub-row **436** define the MRA to be accessed. The state in OP0 in the CA6 bit in sub-row **436** along with the sequence of the states in OP1-OP7 in the CA0-CA6 bits in sub-row **438**, respectively, provide the OP code for the MRPA write operation. Thus, one advantage to the MRPA mode is that the command and the MRP address are not included in the MRPA write extension same page command **412**. The mode register address and the OP code are included in the MRPA write extension same page command **412**. This eliminates two cycles from each additional MRPA write operation that is performed with the MRPA write extension same page command **412**.

[0053] With respect to read operations, the mode register read command **414** can be used for standalone read operations (e.g., when the semiconductor device is not in the MRPA mode). The mode register read command **414** is an example of a direct mode register command. The sequence of the states (H or L) in the CA0-CA4 bits in sub-row **440** define the command to be performed, which is the mode register read command **414**. The sequence of the states in MRP0-MRP4 in the CA0-CA4 bits in sub-row **442** provide the MRP to be accessed. The sequence of the states in MRA0-MRA5 in the CA0-CA5 bits in sub-row **444** define the MRA in the MRP to be accessed.

[0054] The MRPA read same page command **416** can be used to cause the semiconductor device to enter the MRPA mode. In the illustrated embodiment, the external control signal **404** is used to cause the semiconductor device (e.g., cause one or more mode registers in the semiconductor device) to enter the MRPA mode. The external control signal **404** toggles, or is set to, a low state in sub-row **450** (compared to the state of the external control signal **404** in sub-row **444**). The low state in sub-row **450** indicates the semiconductor device is to enter the MRPA mode. Other techniques for causing the semiconductor device to enter the MRPA mode are described in more detail in conjunction with FIGS. 6C-6D.

[0055] In the MRPA read same page command **416**, the sequence of the states (H or L) in the CA0-CA4 bits in sub-row **446** define the command to be performed, which is the MRPA read same page command **416**. The sequence of the states in MRP0-MRP4 in the CA0-CA4 bits in sub-row **448** provides the MRP address to be accessed. The sequence of the states in MRA0-MRA5 in the CA0-

CA5 bits in sub-row **450** define the MRA in the MRP to be accessed.

[0056] When one or more additional MRPA read operations are performed on the same page, the MRPA read extension same page command **418** is used to cause the semiconductor device to remain in the MRPA mode and perform an additional MRPA read operation on the same MRP. The MRPA read extension same page command **418** is received for each additional MRPA read operation to be performed while in the MRPA mode. In the example embodiment, the external control signal **404** toggles, or is set to, a low state in sub-row **452**, which indicates the semiconductor device is to remain in the MRPA mode. The sequence of the states in MRA0-MRA5 in the CA0-CA5 bits in sub-row **452** identify the MRA in the same MRP to be accessed. Thus, another advantage to the MRPA mode is that the command and the MRP address are not included in the MRPA read extension same page command **418**. The MRA is included in the MRPA read extension same page command **418**. This eliminates one cycle from each additional read operation that is performed with the MRPA read extension same page command **418**.

[0057] The command truth table **400** depicts one technique that can be used to exit the MRPA mode. In the MRPA read same page command **416** and in the MRPA read extension same page command **418**, the state of the CA6 bit can be set to a state that indicates the semiconductor device is to exit the MRPA mode (see highlighted area **454**). This example technique is shown and described in more detail in conjunction with FIG. **6D**. In a non-limiting nonexclusive example, the state is a low state, although other embodiments are not limited to this state. Other techniques that may be used to exit the MRPA mode are described in more detail in conjunction with FIG. **6B** and FIG. **6C**.

[0058] FIG. **5** illustrates a second example of a command truth table **500** according to an embodiment of the disclosure. The example command truth table **500** includes an example configuration for a 14-bit CA bus. Although the illustrated command truth table **500** depicts L, H, and V states in some fields, other embodiments are not limited to these states or to the illustrated sequence of states. In some embodiments, the command truth table **500** may be implemented with the semiconductor device **200** of FIG. **2** and/or the memories **110(0)-110(p)** of FIG. **1**. In some embodiments, the command truth table **500** may be implemented with the method **300** of FIG. **3**.

[0059] The command truth table **500** includes column entries for a command (Function) **502**, the external control signal (CS_n) **504**, and the CA bits (CA0-CA13) **506**. The example command truth table **500** includes row entries for each command **502**. The commands **502** include a mode register write command **508**, an MRPA write same page command **510**, an MRPA write extension same page command **512**, a mode register read command **514**, an MRPA read same page command **516**, and an MRPA read extension same page command **518**.

[0060] The mode register write command **508** is a 2-cycle command in which each sub-row **520**, **522** is received at one clock cycle (for a total of two clock cycles for the entire command). The mode register write command **508** is an example mode register write command that is performed at block **306** in FIG. **3**. The MRPA write same page command **510** is a 2-cycle command in which each sub-row **524**, **526** is received at one clock cycle (for a total of two clock cycles for the entire command). The MRPA write extension same page command **512** is a 1-cycle command in which the sub-row **528** is received at one clock cycle. The MRPA write same page command **510** and the MRPA write extension same page command **512** are example MRPA write commands that are performed at block **316** in FIG. **3**.

[0061] The mode register read command **514** is a 2-cycle command in which each sub-row **530**, **532** is received at one clock cycle (for a total of two clock cycles for the entire command). The mode register read command **514** is an example mode register read command that is performed at block **306** in FIG. **3**. The MRPA read same page command **516** is a 2-cycle command in which each sub-row **534**, **536** is received at one clock cycle (for a total of two clock cycles for the entire command). The MRPA read extension same page command **518** is a 1-cycle command in which the sub-row **538** is received at one clock cycle. The MRPA read same page command **516** and the

MRPA read extension same page command **518** are example MRPA read commands that are performed at block **312** in FIG. **3**.

[0062] The mode register write command **508** can be used for standalone write operations (e.g., when the semiconductor device is not in the MRPA mode). The mode register write command **508** is an example of a direct mode register command. The sequence of the states (H or L) in the CA0-CA4 bits in sub-row **520** define the command to be performed, which is the mode register write command **508**. The sequence of the states in MRP0-MRP4 in the CA7-CA11 bits in sub-row **520** provide the MRP address to be accessed. The sequence of the states in MRA0-MRA5 in the CA0-CA5 bits in sub-row **522** identify the MRA in the MRP to be accessed. The sequence of the states in OP0-OP7 in the CA6-CA13 bits in sub-row **522** provide the OP code for the write operation.

[0063] The MRPA write same page command **510** can be used to cause the semiconductor device (e.g., cause one or more mode registers in the semiconductor device) to enter the MRPA mode. In the illustrated embodiment, the external control signal **504** is used to cause the semiconductor device to enter the MRPA mode. The external control signal **504** toggles, or is set to, a low state in sub-row **526** (compared to the state of the external control signal in sub-row **522**), which indicates the semiconductor device is to enter the MRPA mode. This technique is shown and described in more detail in conjunction with FIG. **6A**. Other techniques for causing the semiconductor device to enter the MRPA mode are described in more detail in conjunction with FIGS. **6C-6D**.

[0064] In the MRPA write same page command **510**, the sequence of the states (H or L) in the CA0-CA4 bits in sub-row **524** define the command to be performed, which is the MRPA write same page command **510**. The sequence of the states in MRP0-MRP4 in the CA7-CA11 bits in sub-row **524** provide the MRP address to be accessed. The sequence of the states in MRA0-MRA5 in the CA0-CA5 bits in sub-row **526** define the MRA in the MRP to be accessed. The sequence of the states in OP0-OP7 in the CA6-CA13 bits in sub-row **526** provide the OP code for the write operation.

[0065] When one or more additional MRPA write operations are performed on the same MRP, the MRPA write extension same page command **512** is used to cause the semiconductor device (e.g., cause one or more mode registers in the semiconductor device) to remain in the MRPA mode and perform an additional MRPA write operation on the same page. The MRPA write extension same page command **512** is received for each additional MRPA write operation to be performed while in the MRPA mode. In the example embodiment, the external control signal **504** toggles, or is set to, a low state in sub-row **528**, which indicates the semiconductor device is to remain in the MRPA mode. The sequence of the states in MRA0-MRA5 in the CA0-CA5 bits in sub-row **528** identify the MRA to be accessed. The sequence of the states in OP0-OP7 in the CA6-CA13 bits in sub-row **528** provide the OP code for the MRPA write operation. Thus, another advantage to the MRPA mode is that the command and the MRP address are not included in the MRPA write extension same page command **512**. The MRA and the OP code are included in the MRPA write extension same page command **512**. This eliminates one cycle from each additional MRPA write operation that is performed using the MRPA write extension same page command **512**.

[0066] With respect to read operations, the mode register read command **514** can be used for standalone read operations (e.g., when the semiconductor device is not in the mode register page access mode). The mode register read command **514** is an example of a direct mode register access command. The sequence of the states (H or L) in the CA0-CA4 bits in sub-row **530** define the command to be performed, which is the mode register read command **514**. The sequence of the states in MRP0-MRP4 in the CA7-CA11 bits in sub-row **530** provide the MRP address to be accessed. The sequence of the states in MRA0-MRA5 in the CA0-CA5 bits in sub-row **532** define the MRA in the MRP to be accessed.

[0067] The MRPA read same page command **516** can be used to cause the semiconductor device (e.g., cause one or more mode registers in the semiconductor device) to enter the MRPA mode. In the illustrated embodiment, the external control signal **504** is used to cause the semiconductor

device to enter the MRPA mode. The external control signal **504** toggles, or is set to, a low state in sub-row **536** (compared to the state of the external control signal **504** in sub-row **532**). The low state for the external control signal **504** in the sub-row **536** indicates the semiconductor device is to enter the MRPA mode. Other techniques for causing the semiconductor device to enter the MRPA mode are described in more detail in conjunction with FIGS. **6C-6D**.

[0068] In the MRPA read same page command **516**, the sequence of the states (H or L) in the CA0-CA4 bits in sub-row **534** define the command to be performed, which is the MRPA read same page command **516**. The sequence of the states in MRP0-MRP4 in the CA7-CA11 bits in sub-row **534** provide the MRP address to be accessed. The sequence of the states in MRA0-MRA5 in the CA0-CA5 bits in sub-row **536** define the MRA in the MRP to be accessed.

[0069] When one or more additional MRPA read operations are to be performed on the same page, the MRPA read extension same page command **518** is used to cause the semiconductor device (e.g., cause one or more mode registers in the semiconductor device) to remain in the MRPA mode and perform an additional MRPA read operation on the same page. The MRPA read extension same page command **518** is received for each additional read operation to be performed. In the example embodiment, the external control signal **504** toggles, or is set to, a low state in sub-row **538**, which indicates the semiconductor device is to remain in the MRPA mode. The sequence of the states in MRA0-MRA5 in the CA0-CA5 bits in sub-row **538** define the MRA in the MRP to be accessed. Thus, another advantage to the MRPA mode is that the command and the MRP address are not included in the MRPA read extension same page command **518**. The MRA is included in the MRPA read extension same page command **518**. This eliminates one cycle from each additional read operation that is performed using the MRPA read extension same page command **518**.

[0070] The example command truth table **500** depicts the technique of using a state of a CA bit (e.g., the CA6 bit) to indicate the semiconductor device (e.g., one or more mode registers in the semiconductor device) is to exit the MRPA mode (see highlighted area **540**). This example technique is shown and described in more detail in conjunction with FIG. **6D**. In a non-limiting nonexclusive example, the state is a low state, although other embodiments are not limited to this state. Other techniques that may be used to cause the semiconductor device to exit the MRPA mode are described in more detail in conjunction with FIG. **6B** and FIG. **6C**.

[0071] FIG. **6A** illustrates a first technique for entering and remaining in the MRPA mode according to an embodiment of the disclosure. The external control signal **600** can be set to a particular state, such as a low (L) state **602**, in the MRPA write same page command, the MRPA write extension same page command, the MRPA read same page command, and/or the MRPA read extension same page command. Examples of the use of the state of the external control signal are shown and described in conjunction with the MRPA write same page commands, the MRPA write extension same page commands, the MRPA read same page commands, and/or the MRPA read extension same page commands in FIG. **4** and FIG. **5**.

[0072] FIG. **6B** illustrates a first technique for exiting the MRPA mode according to an embodiment of the disclosure. The external control signal **600** can be set to a particular state, such as a low (L) state, for multiple consecutive clock cycles in the MRPA write same page command, the MRPA write extension same page command, the MRPA read same page command, and/or in the MRPA read extension same page command. In FIG. **6B**, the external control signal **600** is set to a low state in two consecutive clock cycles **604**, **605** (e.g., two consecutive rising edges or falling edges). In a non-limiting nonexclusive example, the state of the external control signal in sub-row **438** (FIG. **4**) can toggle to the L state (instead of the H state shown in FIG. **4**) to indicate the semiconductor device is to exit the MRPA mode.

[0073] FIG. **6C** illustrates a second technique for entering and/or exiting the MRPA mode according to an embodiment of the disclosure. The MRA bits (e.g., MRA0-MRA5 **606**) can all be set to one state, such as a H (e.g., "1") state **608**, to cause the semiconductor device (e.g., cause one or more mode registers in the semiconductor device) to enter the MRPA mode. In one embodiment

of the disclosure, the state of the external control signal may be used to cause the semiconductor device to remain in the MRPA mode. To exit the MRPA mode, the MRA bits may all be set to the opposite state, such as an L (e.g., "0") state **610**. This technique can be used in the MRPA write same page command, the MRPA write extension same page command, the MRPA read same page command, and/or in the MRPA read extension same page command.

[0074] FIG. 6D illustrates a third technique for entering and/or exiting the MRPA mode according to an embodiment of the disclosure. A CA_n bit **612** can be set to a particular state, such as an L state **614**, to cause the semiconductor device (e.g., cause one or more mode registers in the semiconductor device) to enter the MRPA mode. In some embodiments of the disclosure, the state of the CA_n bit **612** can remain in the same state to cause the semiconductor device to remain in the MRPA mode, or the state of the external control signal may be used to cause the semiconductor device to remain in the MRPA mode. To exit the MRPA mode, the CA_n bit may be set the opposite state, such as an H state **616**. This technique can be used in the MRPA write same page command, the MRPA write extension same page command, the MRPA read same page command, and/or in the MRPA read extension same page command.

[0075] It is to be appreciated that any one of the examples, embodiments or processes described herein may be combined with one or more other examples, embodiments and/or processes or be separated and/or performed amongst separate devices or device portions in accordance with the present systems, devices and methods. For example, the various techniques for enter and exiting the MRPA mode may be combined. One embodiment of combined techniques is shown in FIG. 4 and FIG. 5 where the external control signal is used to cause the semiconductor device to enter the MRPA mode in both the mode register write same page and the mode register read same page, and the CA bit is used to cause the mode register to exit the MRPA mode. Additionally, although embodiments are described with the semiconductor device entering and exiting the MRPA mode, other embodiments are not limited to this implementation. The semiconductor device may perform operations without entering into a mode.

[0076] Finally, the above discussion is intended to be merely illustrative of the present system and should not be construed as limiting the appended claims to any particular embodiment or group of embodiments. Thus, while the present system has been described in particular detail with reference to exemplary embodiments, it should also be appreciated that numerous modifications and alternative embodiments may be devised by those having ordinary skill in the art without departing from the broader and intended spirit and scope of the present system as set forth in the claims that follow. Accordingly, the specification and drawings are to be regarded in an illustrative manner and are not intended to limit the scope of the appended claims.

Claims

1. An apparatus, comprising: external terminals configured to receive a plurality of commands associated with an operation, the operation comprising a mode register read operation or a mode register write operation; and one or more mode registers each configured to perform the operation, wherein: a first command in the plurality of commands causes the apparatus to enter a mode register page access mode to enable the one or more registers to be accessed as a page, the first command providing a mode register page address and a mode register address in the mode register page to access; and a second command in the plurality of commands causes the apparatus to remain in the mode register page access mode and provides a subsequent mode register address in the mode register page to access.
2. The apparatus of claim 1, wherein a third command in the plurality of commands causes the apparatus to exit the mode register page access mode.
3. The apparatus of claim 2, wherein the third command comprises an external control signal set at a particular state to cause the apparatus to exit the mode register page access mode.

4. The apparatus of claim 1, wherein the first command and the second command are mode register page access commands that cause the apparatus to perform a mode register page access operation.
5. The apparatus of claim 1, wherein a fourth command in the plurality of commands is a direct mode register access command that causes the apparatus to perform a direct mode register access operation.
6. The apparatus of claim 1, wherein the first command comprises an external control signal set at a particular state to cause the apparatus to enter the mode register page access mode.
7. The apparatus of claim 1, wherein the second command comprises an external control signal set at a particular state to cause the apparatus to remain in the mode register page access mode.
8. The apparatus of claim 1, wherein the first command comprises one or more command address bits each set at a particular state to cause the apparatus to enter the mode register page access mode.
9. The apparatus of claim 8, wherein the one or more command address bits comprise multiple command address bits that define a mode register address.
10. The apparatus of claim 1, wherein the second command comprises one or more command address bits each set at a particular state to cause the apparatus to exit the mode register page access mode.
11. The apparatus of claim 10, wherein the one or more command address bits comprise multiple command address bits that define a mode register address.
12. The apparatus of claim 1, wherein the one or more mode registers comprise two or more mode registers that are associated with a similar function.
13. A method, comprising: receiving a first command to perform a first operation of a first type, the first type comprising a mode register read operation or a mode register write operation; based on the first command, causing a mode register to enter a mode register page access mode; receiving a second command to perform a second operation of the first type; and based on the second command, causing the mode register to remain in the mode register page access mode, wherein: the first command provides a mode register page address and a first mode register address in the mode register page to access; and the second command provides a second mode register address in the mode register page to access.
14. The method of claim 13, further comprising receiving a third command to cause the mode register to exit the mode register page access mode.
15. The method of claim 14, wherein the third command comprises an external control signal set at a particular state to cause the mode register to exit the mode register page access mode.
16. The method of claim 13, wherein: the first command comprises an external control signal set at a particular state to cause the mode register to enter the mode register page access mode; or the second command comprises an external control signal set at a particular state to cause the mode register to remain in the mode register page access mode.
17. The method of claim 13, wherein: the first command comprises one or more command address bits each set at a particular state to cause the mode register to enter the mode register page access mode; or the second command comprises one or more command address bits each set at a particular state to cause the mode register to exit the mode register page access mode.
18. The method of claim 13, wherein: the first command comprises a plurality of mode register address bits each set at a particular state to cause the mode register to enter the mode register page access mode; or the second command comprises a plurality of mode register address bits each set at a particular state to cause the mode register to exit the mode register page access mode.
19. A controller configured to transmit commands to a memory device, the commands comprising: a first command to cause the memory device to enter a mode register page access mode; and a second command to cause the memory device to remain in the mode register page access mode, wherein: the first command comprises a first set of bits: at least one bit in the first set of bits causes the memory device to enter the mode register page access mode; a first portion of the first set of bits provides a mode register page address; a second portion of the first set of bits provides a mode

register address in the mode register page to access; the second command comprises a second set of bits: at least one bit in the second set of bits causes the memory device to remain in the mode register page access mode; and a third portion of the second set of bits provides a mode register address in the mode register page to access.

20. The apparatus of claim 19, wherein the at least one bit in the first set of bits is an external control signal.

21. The apparatus of claim 19, wherein the at least one bit in the second set of bits is an external control signal.

22. The apparatus of claim 19, wherein the at least one bit in the first set of bits comprises one or more command address bits.

23. The apparatus of claim 19, wherein the at least one bit in the second set of bits comprises one or more command address bits.
